Agentic RAG: Tool Routing Accuracy Under Prompt Ambiguity

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1. Background

- Retrieval Augmented Generation (RAG) enhances LLM reliability by grounding responses in retrieved documents. Traditional RAG uses a fixed retrieval source, but modern agentic systems must dynamically choose between tools such as a local vector store or a live web search API.
- Tool routing is a critical reliability bottleneck: routing a time-sensitive query to static document store produces stale answers, while routing a document-specific query to web search produces irrelevant, off-topic results.
- Despite growing adoption of agentic RAG, little work systematically evaluates how routing accuracy degrades

2. Motivations & Objectives

Routing errors in agentic LLM systems lead to responses that are outdated, off-topic, or confidently wrong yet most existing benchmarks evaluate final answer quality rather than the routing decision itself. This project isolates routing as the primary variable.

- Build an agentic RAG system (LangGraph + ChromaDB + Tavily) routing financial queries between a local vector store of 22 corporate 10-K filings and live web search
- Design and compare 6 routing strategies: always-local baseline, rule-based keyword matching, zero-shot LLM, few-shot LLM, chain-of-thought LLM, and a full LangGraph stateful agent
- Evaluate across 4 LLMs: Claude Sonnet 4.6, GPT-4o-mini, Gemma-3-27B, LLaMA-3-8B via OpenRouter API
- Measure routing accuracy degradation across 3 prompt ambiguity tiers (clear → moderate → highly ambiguous)
- Identify how routing is affected by prompt ambiguity and which prompt engineering strategy produces the most reliable router and pinpoint linguistic failure patterns

3. Related Work

RAG Systems: Lewis et al. (2020) introduced RAG to augment LLMs with non-parametric retrieval, improving factual grounding. Subsequent work extended this to multi-hop and open-domain settings.

Agentic LLMs & Tool Use: ReAct (Yao et al., 2023) and Toolformer (Schick et al., 2023) demonstrated that LLMs can reason about and invoke external tools. LangGraph formalizes this into stateful, conditional agent graphs.

Prompt Engineering: Wei et al. (2022) showed chain-of-thought prompting improves complex reasoning. Brown et al. (2020) established few-shot in-context learning as a powerful alternative to fine-tuning.

Gap: No prior work constructs a labeled dataset explicitly designed to stress-test routing decisions across a continuous ambiguity spectrum with matched ground-truth tool labels.

4. Methodology

Dataset:

The project has 100 prompts which are drawn from 22 tech company FY2025 10-K filings (Amazon, Apple, Alphabet, Microsoft, Adobe, Netflix, Uber, Airbnb, Visa, Snowflake, and more).

Each prompt is labelled with a ground-truth tool (local or web) and assigned to one of three ambiguity tiers:

Table 1 - Dataset

Tier	Label	Example Prompt
Tier 1 - Unambiguous	40% of dataset	"What was Amazon's FY2025 net income?"
Tier 2 - Moderately ambiguous	30% of dataset	"What are the main risks Airbnb faces?"
Tier 3 - Highly ambiguous	30% of dataset	"Is Amazon a good long-term investment?"

System Architecture:

Flowchart for diagram for query routing system

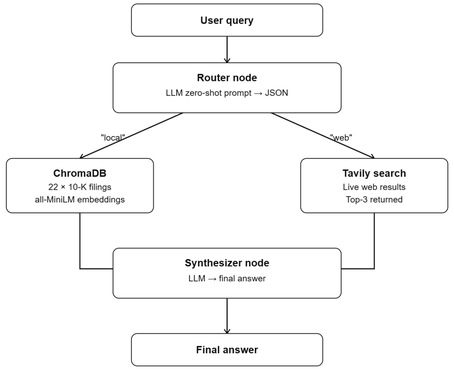

Fig. 1 - LangGraph agentic pipeline

Routers Evaluated:

1. Always-Local - routes every query to ChromaDB regardless of content (floor baseline)
2. Rule-Based - keyword matching: detects "today", "current", "10-K" etc., returns "uncertain" if no match found
3. Zero-Shot LLM - prompt: classify as local or web, output strict JSON {"tool": ..., "reason": ...}, no examples
4. Few-Shot LLM - same as zero-shot + 3 labelled examples (1 local, 1 web, 1 ambiguous-web) in prompt
5. Chain-of-Thought (CoT) - 3-step: check fiscal year reference → check temporal keywords → resolve ambiguity
6. LangGraph Agent - zero-shot router embedded in stateful graph, uncertain decisions fall back to local retrieval

Models & Config:

Anthropic Claude Sonnet 4.6 | Google Gemma 3 27B | GPT-4o-mini | Llama 3 8B

Team contribution:

Common - Poster and presentation
Henil Patel - System architecture, LangGraph agent pipeline, Evaluation framework
Parin Shah - Dataset curation, Prompt design across ambiguity tiers, Routing baseline implementation.
Heta Shah - Dataset curation, Routing strategies, Results visualization

5. Experimental Results

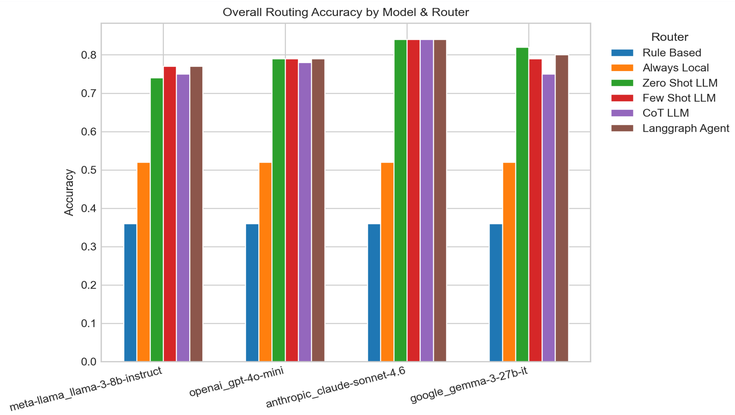

Fig. 2 - Overall routing accuracy per model and strategy

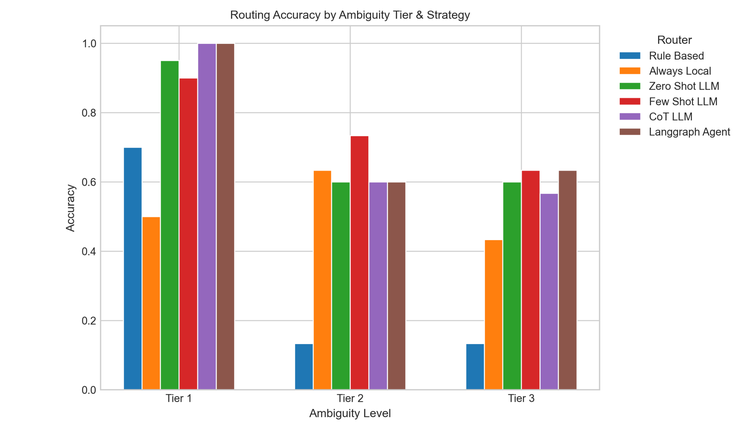

Fig. 3 - Accuracy by ambiguity tier

Key Findings:

LLM-based routers achieve high accuracy on clear queries but routing accuracy degrades significantly as query ambiguity increases, dropping from 100% (Tier 1) to 73% (Tier 3) even for the strongest model (Sonnet 4.6). This demonstrates that query ambiguity, not model size or prompt strategy, is the primary bottleneck in agentic RAG routing.

Sonnet 4.6 achieves the highest accuracy (84%) across all LLM-based strategies, followed by Gemma-3-27B (82%), GPT-4o-mini (79%), and LLaMA-3-8B (76%). Rule-based accuracy remains flat at 36% across all models.

All strategies degrade sharply from Tier 1 to Tier 2, with zero-shot dropping from 95% to 60% and few-shot from 90% to 73%. Rule-based collapses entirely at Tiers 2 and 3 (13%).

Model capability matters more than prompt strategy. Claude achieves 84% across all 3 prompt versions, adding few-shot or CoT changes nothing for strong models.

Future work: Confidence-based fallback via cosine similarity threshold, answer quality via LLM-as-judge, expanding testing the routing strategies on different types of datasets